

Sampling and population

Dr Patricia Ryser-Welch, June 2025

***How did Pierre Simon Laplace
estimated the population of
France in 1783?***

He applied a double sampling approach where he used two available samples of the population of France consisting of birth records (list *A*) and available regional census data (list *B*). Then, each individual observed in the study falls into exactly one of three categories:

- Observed on both lists *A* and *B*.
- Observed on list *A* but not observed on list *B*.
- Observed on list *B* but not observed on list *A*.

The data are summarised as the number of individuals that fall into each of these three categories. The true population size, however, is not just the individuals observed in the study since this fails to include those individuals that were not observed on either list *A* or list *B*. The number of individuals not observed on either list is called the hidden or dark figure.

What is the estimated population of Hogwarts?

Pupils appearing in both books

	Philosopher stones	Deathly Hallows
0	Harry Potter	Neville LongBottom
1	Hermione Granger	Ginny Weasley
2	Ron Weasley	Seamus Finnigan
3	Neville Longbottom	Dean Thomas
4	Seamus Finnigan	Lavender Brown
5	Dean Thomas	Parvati Patil
6	Lavender Brown	Colin Creevey
7	Parvati Patil	Dennis Creevey
8	Draco Malfoy	Lee Jordan
9	Vincent Crabbe	Ernie Macmillan
10	Gregory Goyle	Hannah Abbott
11	Pansy Parkinson	Justin Finch-Fletchley
12	Millicent Bulstrode	Susan Bones
13	Theodore Nott	Zacharias Smith
14	Blaise Zabini	Luna Lovegood
15	Daphne Greengrass	Padma Patil:
16	Hannah Abbott	Cho Chang
17	Susan Bones	Anthony Goldstein
18	Justin Finch-Fletchley	Michael Corner
19	Ernie Macmillan	Terry Boot
20	Padma Patil	Draco Malfoy
21	Terry Boot	Gregory Goyle
22	Mandy Brocklehurst	Vincent Crabbe
23	Anthony Goldstein	Pansy Parkinson:
24	Michael Corner	Blaise Zabini
25	Lisa Turpin	Theodore Nott
26	Morag MacDougal	Daphne Greengrass

```
{'ANTHONY GOLDSTEIN',  
  'BLAISE ZABINI',  
  'DAPHNE GREENGRASS',  
  'DEAN THOMAS',  
  'DRACO MALFOY',  
  'ERNIE MACMILLAN',  
  'GREGORY GOYLE',  
  'HANNAH ABBOTT',  
  'JUSTIN FINCH-FLETCHLEY',  
  'LAVENDER BROWN',  
  'MICHAEL CORNER',  
  'NEVILLE LONGBOTTOM',  
  'PARVATI PATIL',  
  'SEAMUS FINNIGAN',  
  'SUSAN BONES',  
  'TERRY BOOT',  
  'THEODORE NOTT',  
  'VINCENT CRABBE'}
```

Lincoln-Peterson estimator

Data:

U = total number of individuals observed on list A;

V = total number of individuals observed on list B;

W = number of individuals observed on list A and list B.

Lincoln-Peterson estimator for total population, N , is:

$$\hat{N} = \frac{UV}{W}$$

Let's do the calculations

Estimation of the population:

U is the total number of pupils in the first book

V is the total number of pupils in the last book

W is the number of pupils observed in both books

$$\hat{N} = \frac{UV}{W}$$

```
U = len(year_1)
U
```

27

+ Code

+ Markdown

```
V = len(year_6)
V
```

27

```
W = count_both
W
```

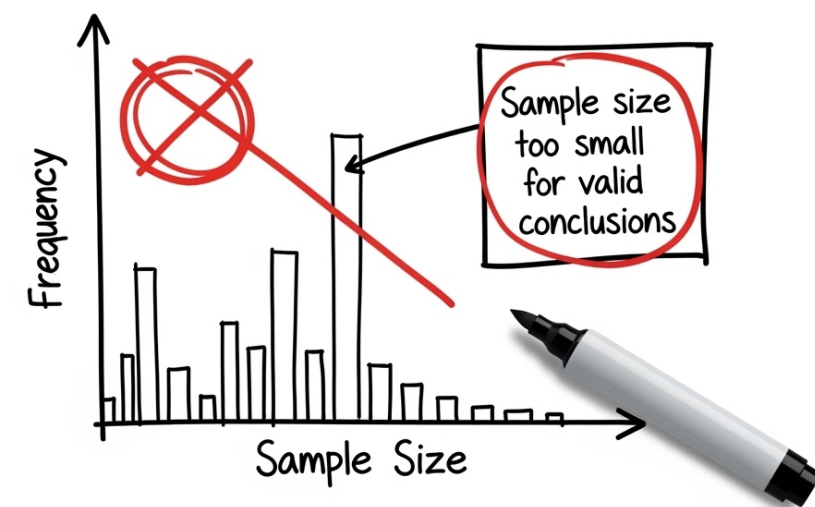
18

```
N = (U * V) / W
N
```

40.5

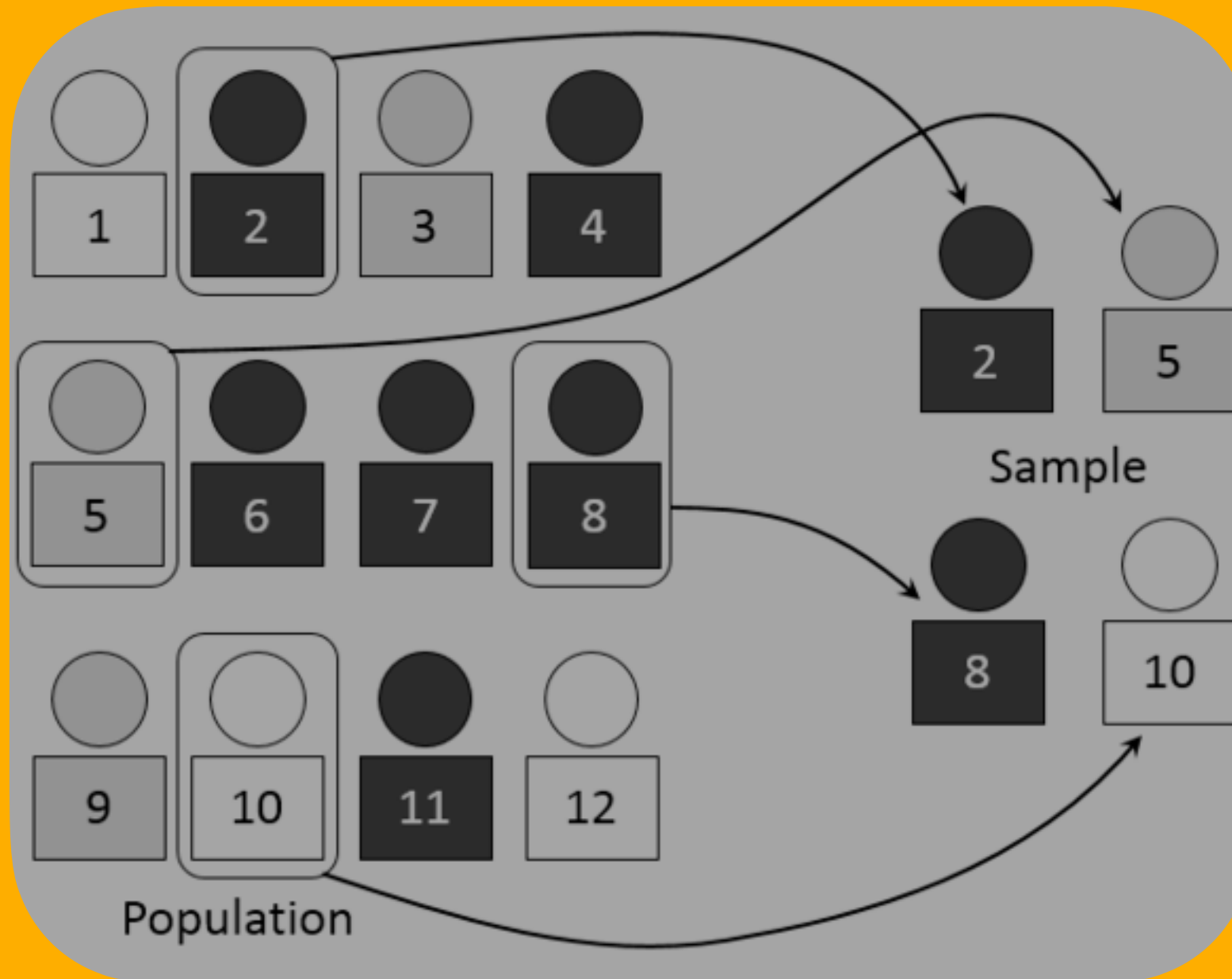
This estimation is not quite accurate...

It has been estimated as a minimum as a 1,000 pupils.



Population and sample

Population vs Sample



By Dan Kernler - Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=36506020>

Population and sample mean

Population Mean	Sample Mean
$\mu = \frac{\sum_{i=1}^N x_i}{N}$ N = number of items in the population	$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$ n = number of items in the sample

Population vs sample standard deviation

Standard Deviation Formula ungrouped data

Population

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

σ - Population standard deviation

$\sum_{i=1}^N$ - Summation of all values (1-N)

x_i - Each value in the population

μ - Population mean

N - Population size

Sample

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

s - Sample standard deviation

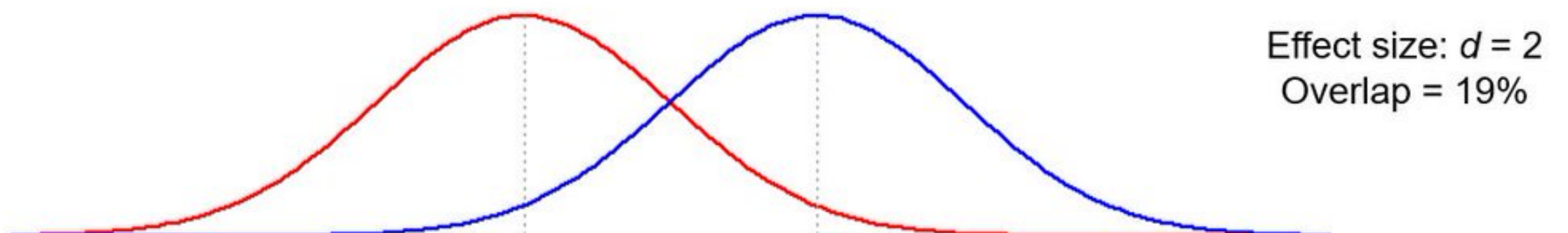
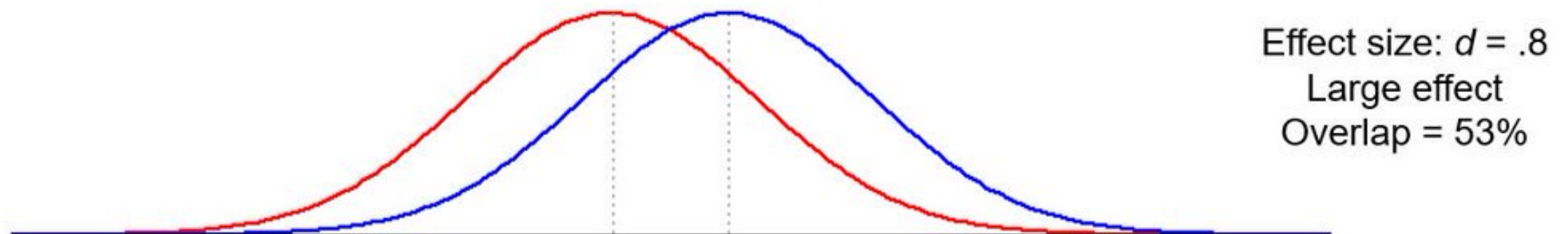
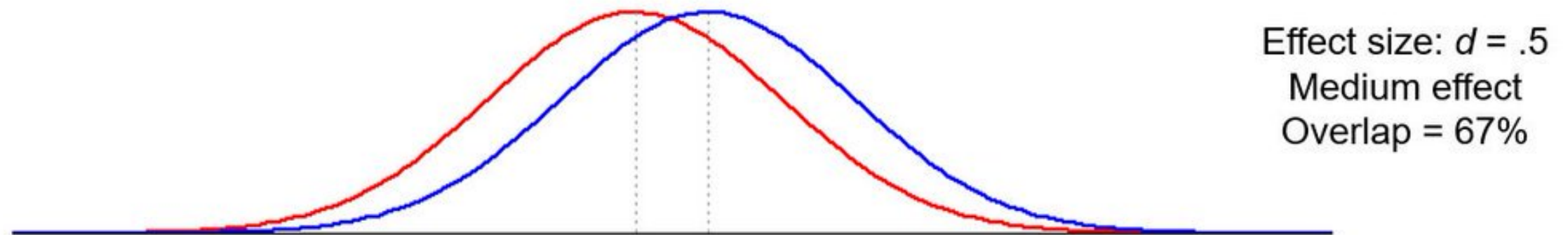
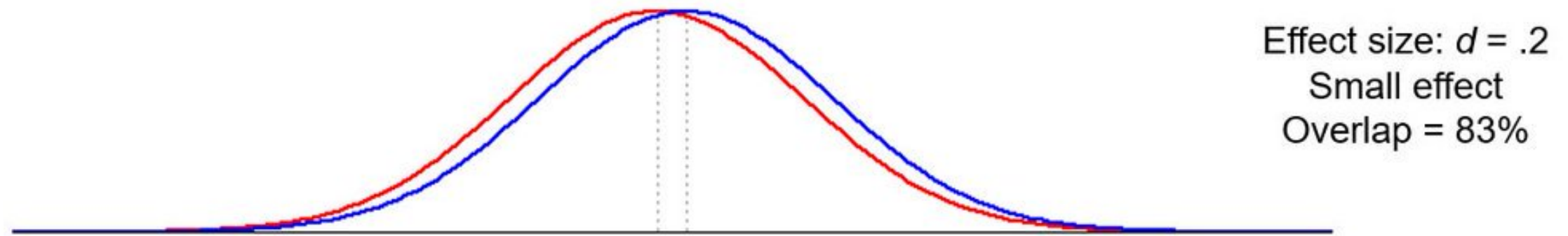
$\sum_{i=1}^n$ - Summation of all values (1-n)

x_i - Each value in the sample

$\bar{x}$ - Sample mean

n - Sample size

Understanding Effect Sizes



Measuring effect size with Cohen's d

Cohen's d

Cohen's d is probably the most popular effect size to use in hypothesis testing for continuous data, where the sample size is large (more than 50), and is calculated as:

$$d = \frac{\text{mean of group 1} - \text{mean of group 2}}{\text{pooled standard deviation}}$$

Generally:

- $d \leq 0.5$ is a small effect size
- $0.5 < d < 0.8$ is a medium effect size
- $0.8 \leq d < 1.3$ is a large effect size
- $1.3 \leq d$ is a very large effect size.

Notice that the equation requires the pooled standard deviation - this is of course assuming equal variance between groups! If you are not assuming this, you will need to use an alternative, such as Glass's Δ . The pooled standard deviation is a weighted average of the two group standard deviations, calculated by:

$$\text{pooled standard deviation} = \sqrt{\frac{s_1^2 + s_2^2}{2}}$$

where

- n_1 is the sample size for group 1
- n_2 is the sample size for group 2
- s_1 is the standard deviation for group 1
- s_2 is the standard deviation for group 2.

How to interpret the effect size?

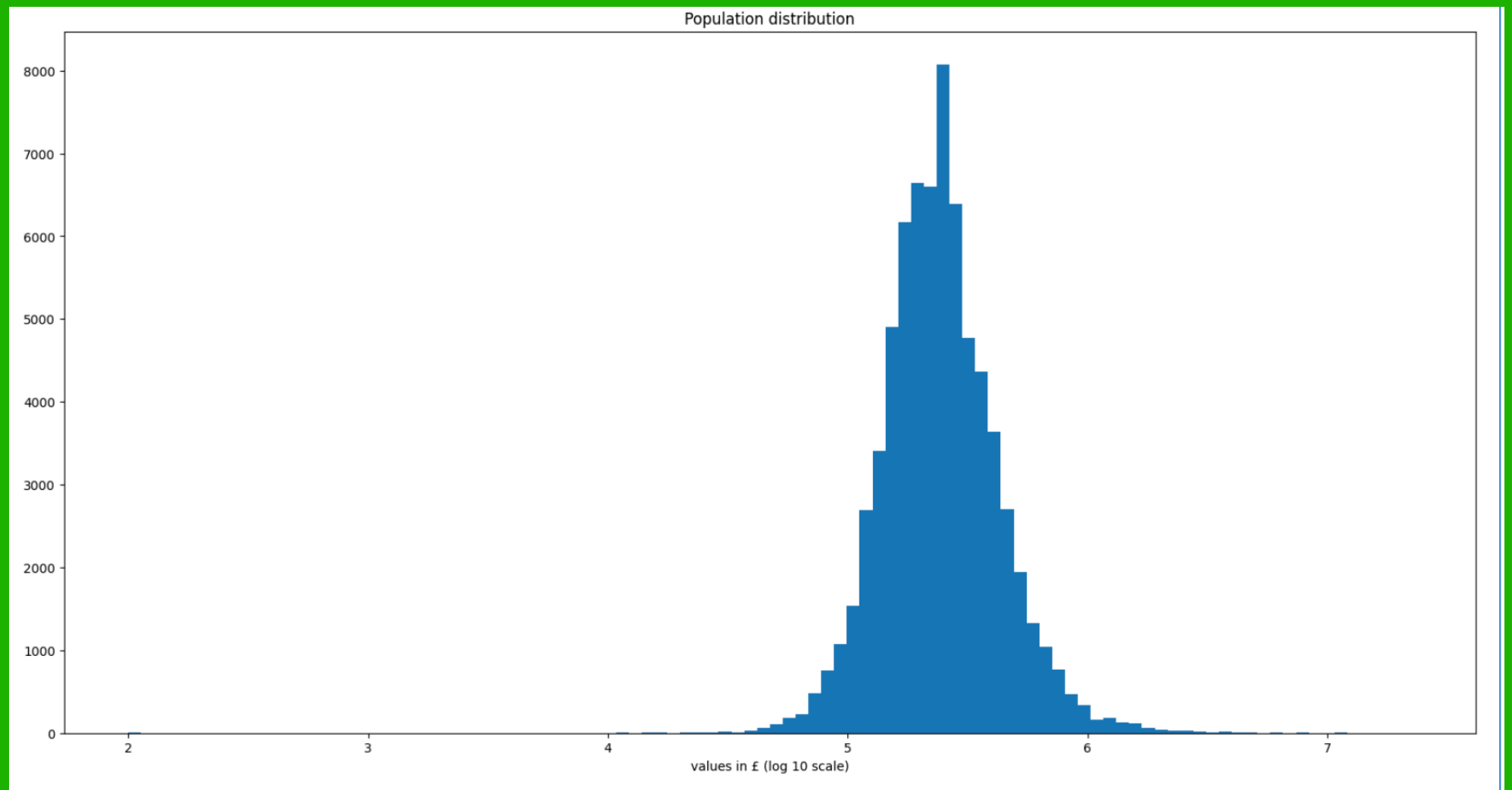
- Cohen's d (for comparing two means): [↗](#)
 - Formula: $d = (\text{Mean}_1 - \text{Mean}_2) / \text{Standard Deviation}$ (often a pooled standard deviation) [↗](#)
 - Interpretation: It represents the difference between two means in terms of standard deviation units. [↗](#)
 - General Guidelines (Cohen's conventions, though context-dependent):
 - 0.2: Small effect (the means differ by 0.2 standard deviations) [↗](#)
 - 0.5: Medium effect (the means differ by 0.5 standard deviations) [↗](#)
 - 0.8: Large effect (the means differ by 0.8 standard deviations or more) [↗](#)
 - Visualizing: A d of 0.8 means that the mean of the "treatment" group is at the 79th percentile of the "control" group.

An example

Property prices Cambridge

Prices between 1/01/2001 and 29/06/2017

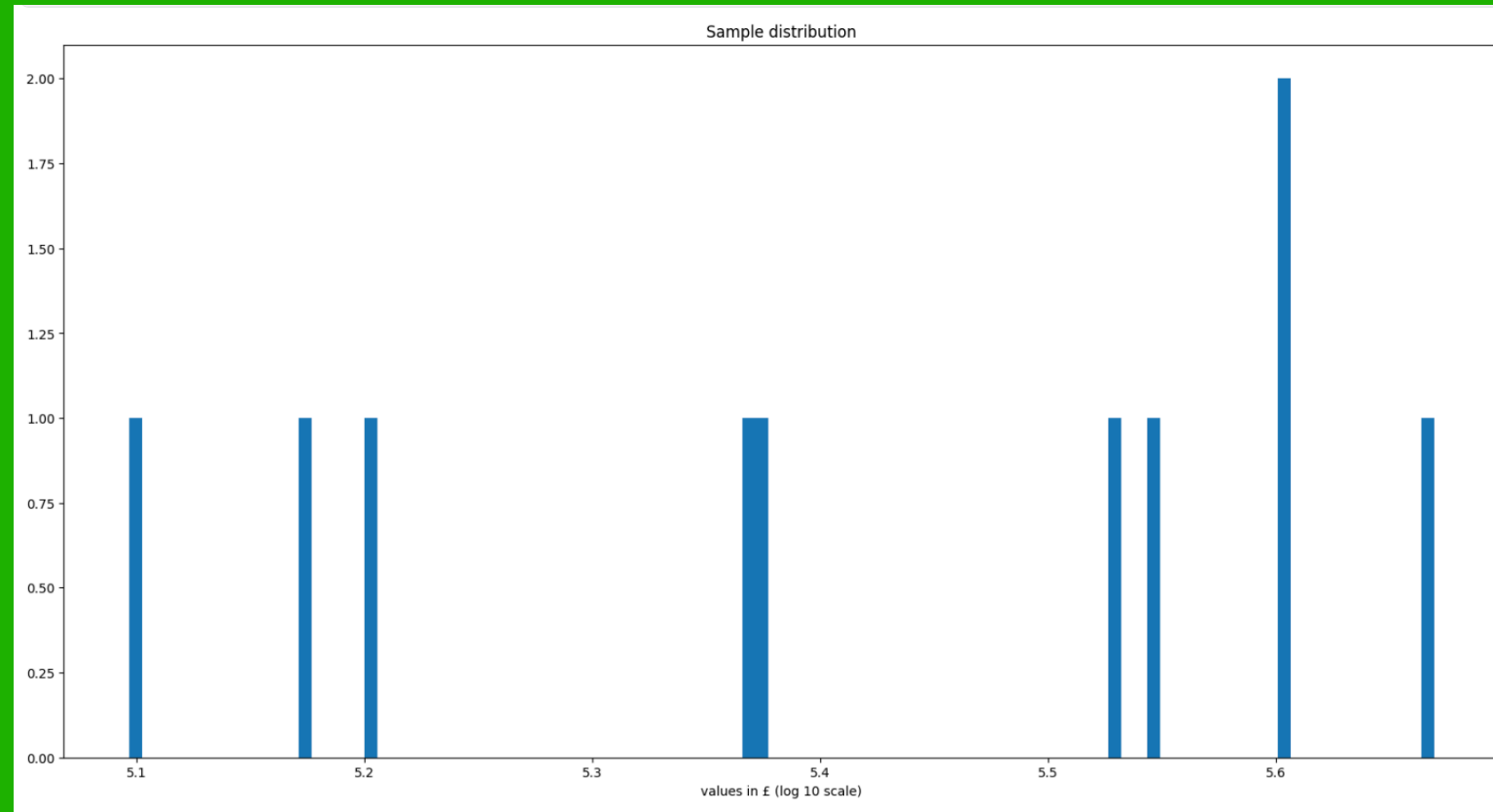
Price	
count	7.158900e+04
mean	2.887456e+05
std	2.916205e+05
min	1.000000e+02
25%	1.700000e+05
50%	2.360000e+05
75%	3.350000e+05
max	2.237400e+07



The distribution is in logarithm scale - log 10.

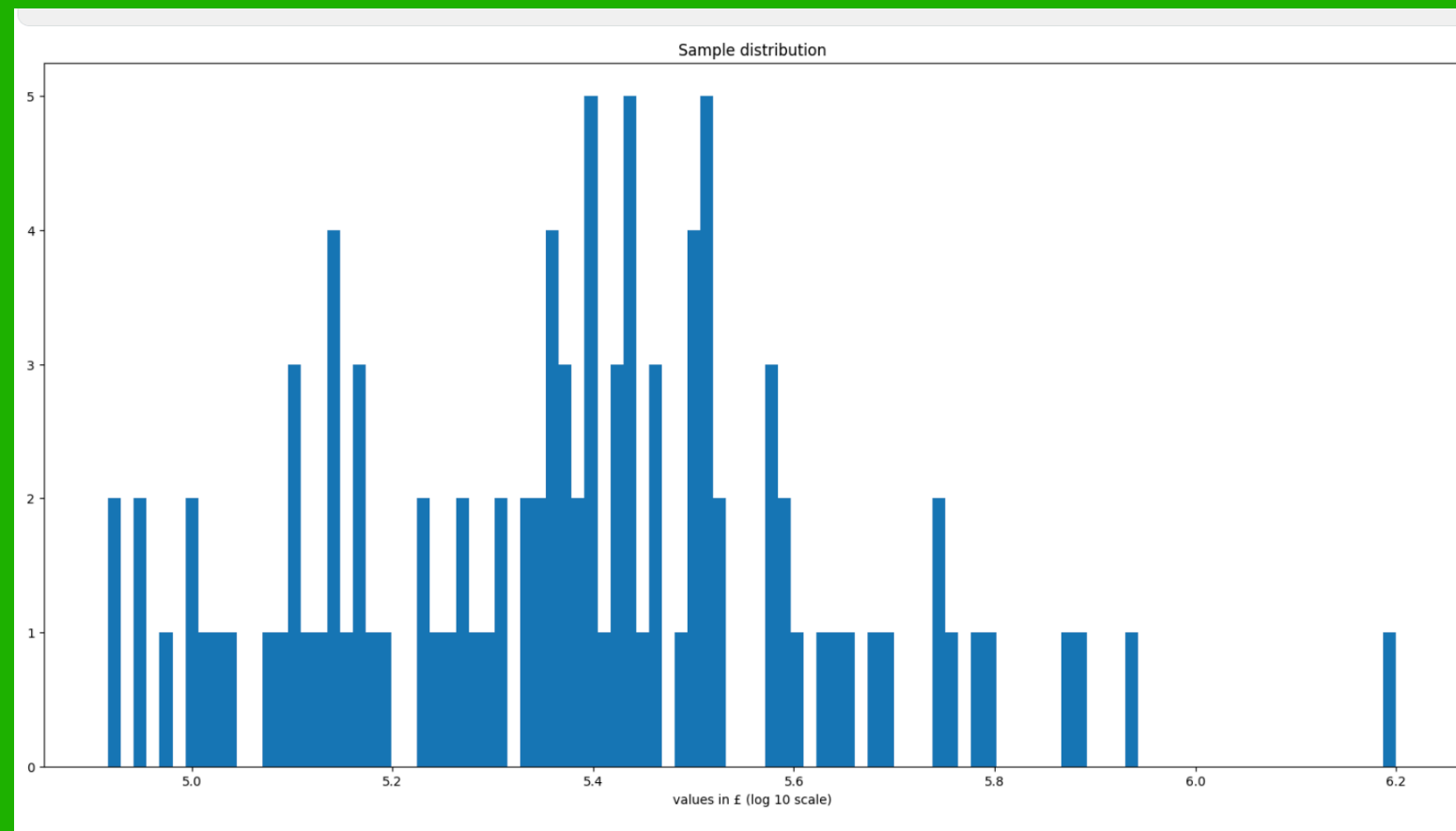
Sample size : 10

Population mean : 288745.5937364679
Population standard deviation : 291620.51628526766
Sample mean : 286720.0
Sample standard deviation : 120864.78027577393



Sample size : 100

Population mean : 288745.5937364679
Population standard deviation : 291620.51628526766
Sample mean : 283157.57
Sample standard deviation : 201691.93464858184



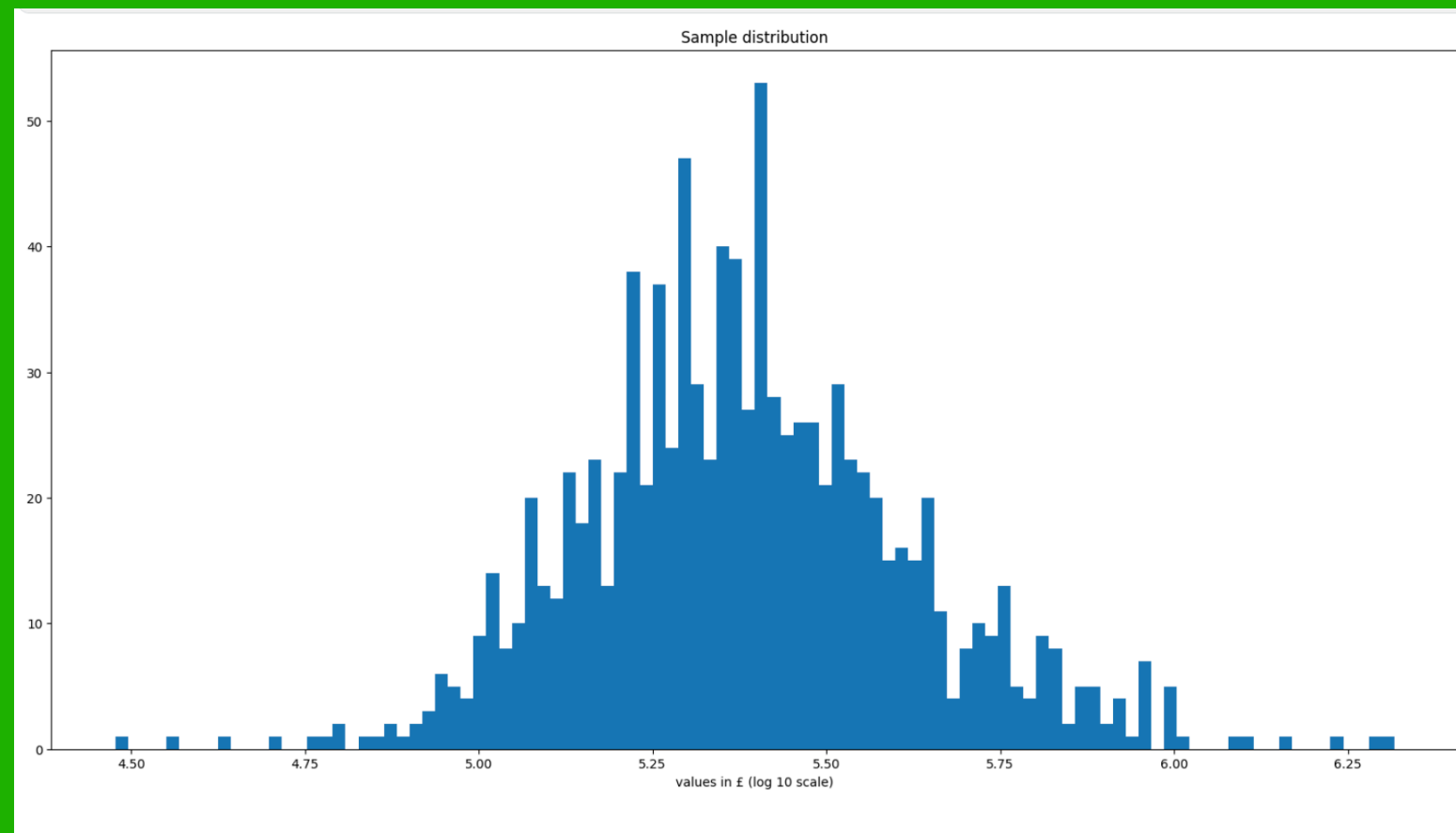
Sample size : 1000

Population mean : 288745.5937364679

Population standard deviation : 291620.51628526766

Sample mean : 285159.11

Sample standard deviation : 194912.58219439737



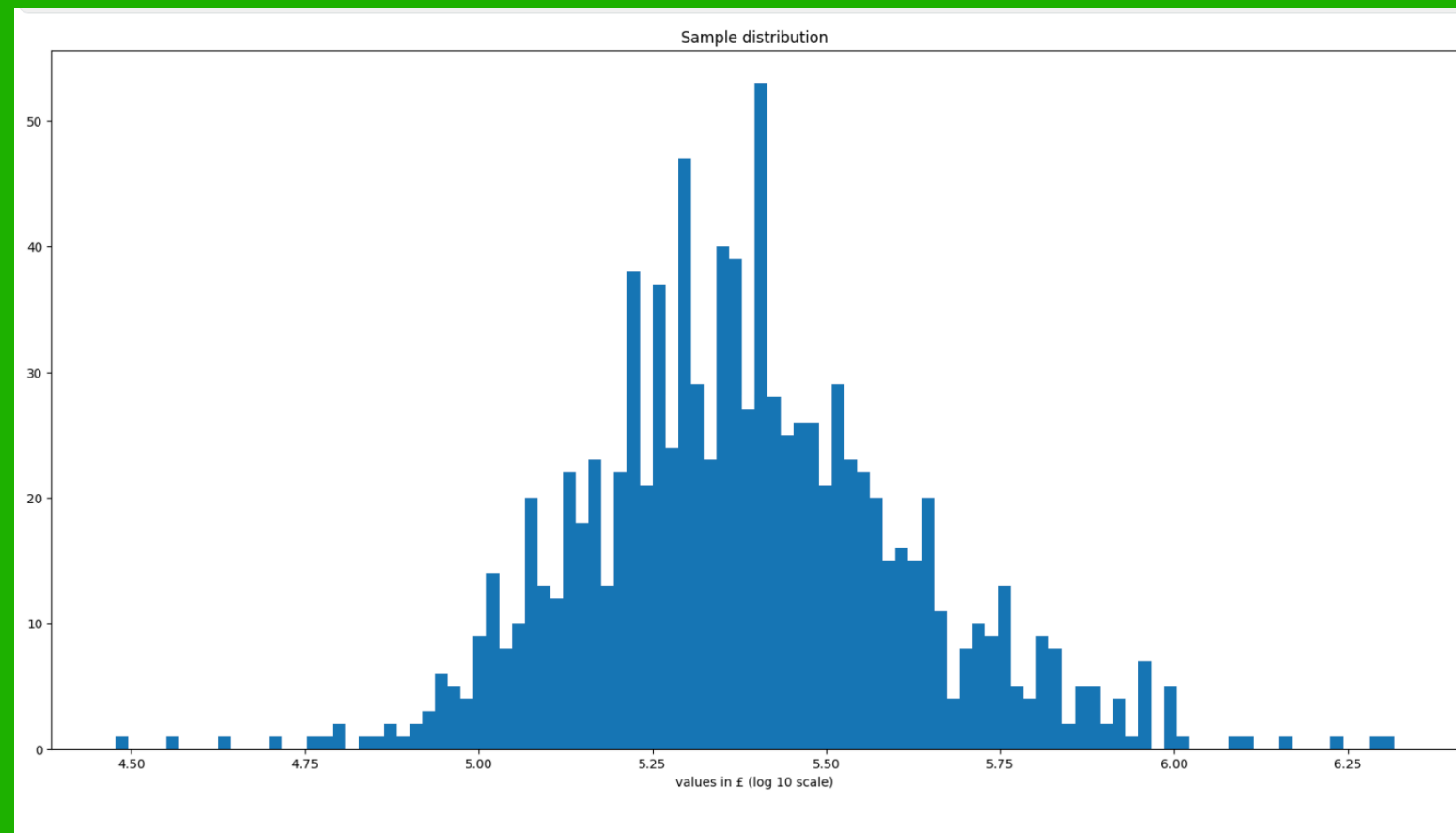
Sample size : 1000

Population mean : 288745.5937364679

Population standard deviation : 291620.51628526766

Sample mean : 285159.11

Sample standard deviation : 194912.58219439737



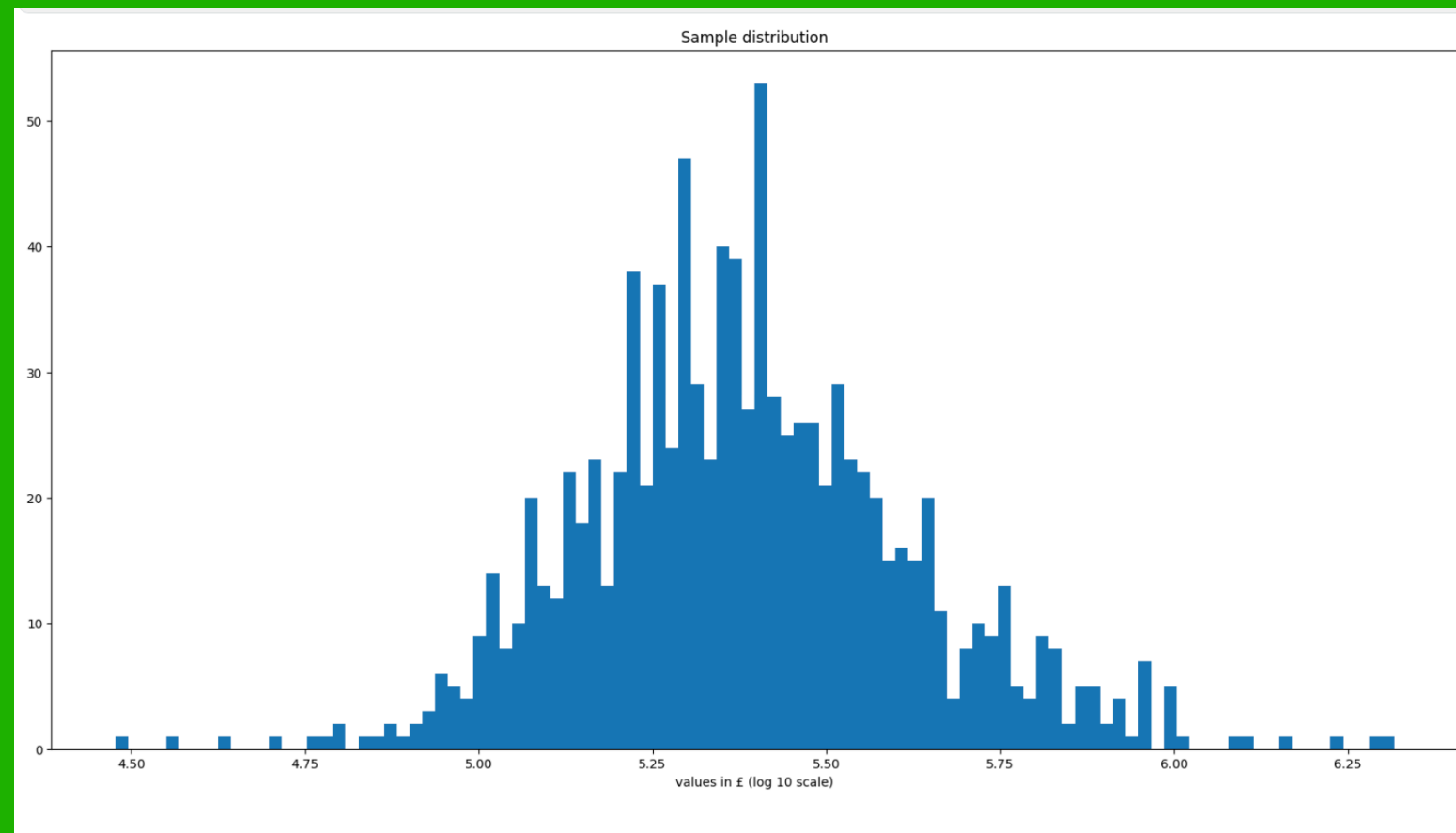
Sample size : 1000

Population mean : 288745.5937364679

Population standard deviation : 291620.51628526766

Sample mean : 285159.11

Sample standard deviation : 194912.58219439737



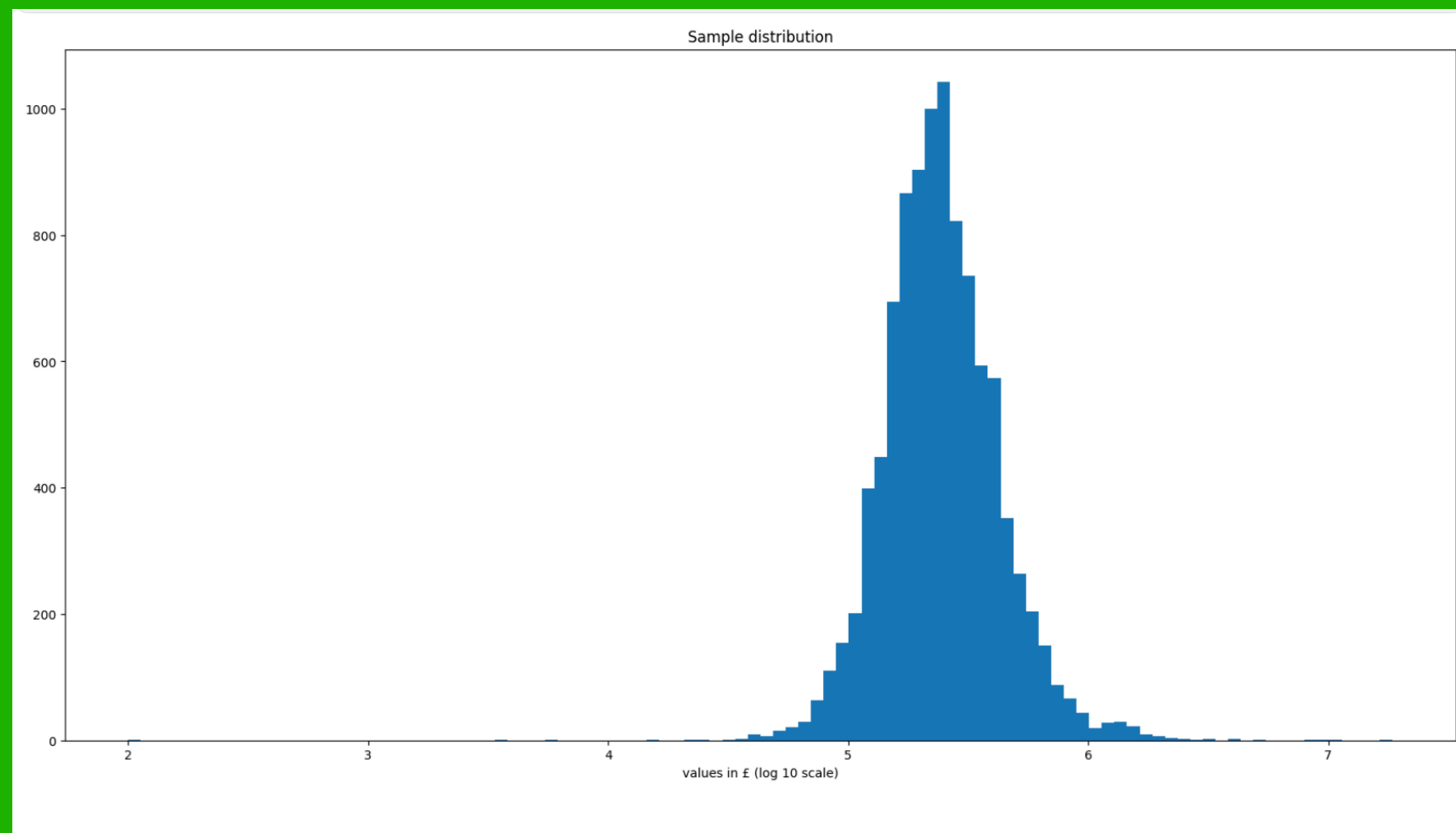
Sample size : 10000

Population mean : 288745.5937364679

Population standard deviation : 291620.51628526766

Sample mean : 291576.3608

Sample standard deviation : 329796.13268337393



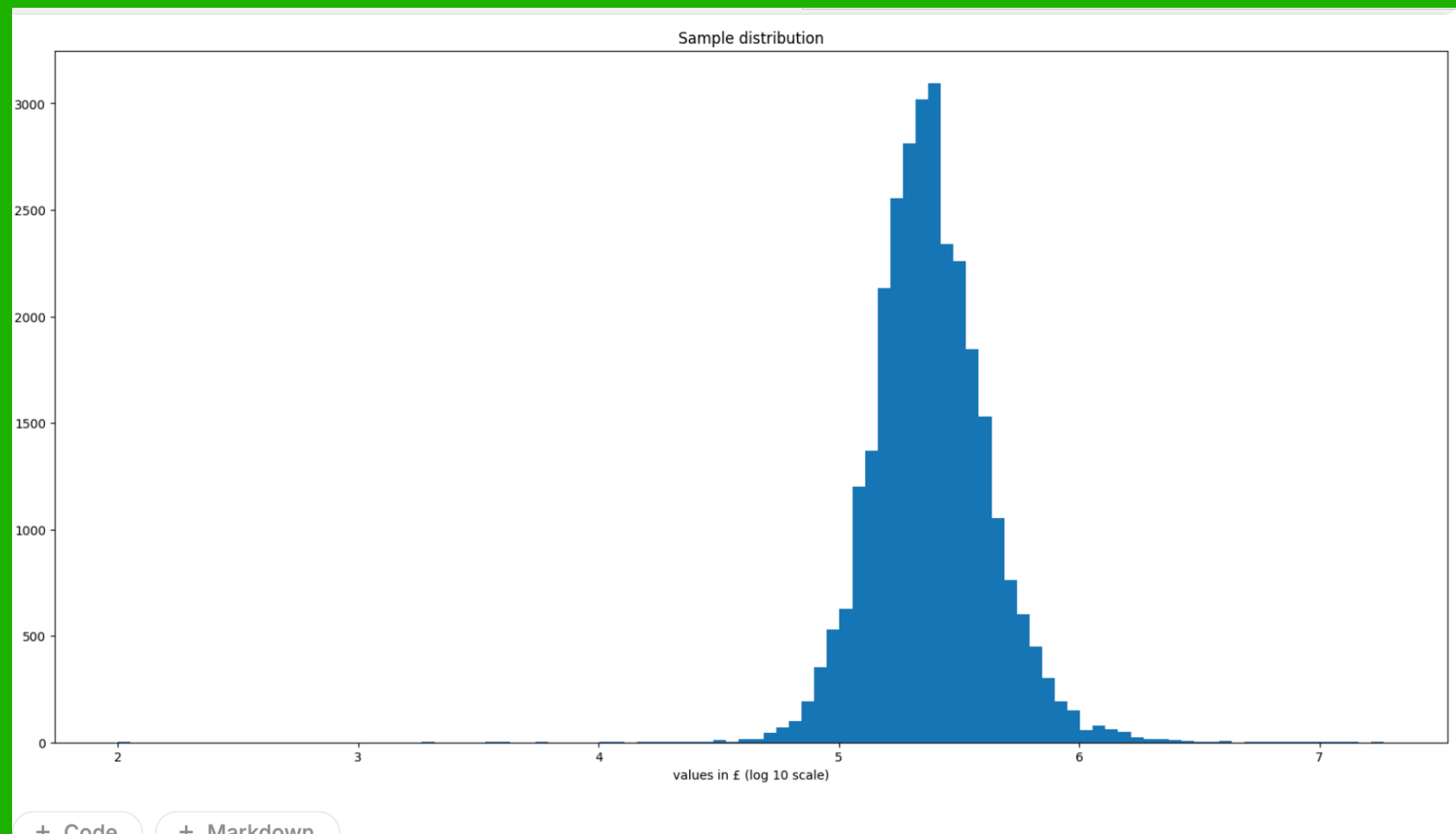
Sample size : 30000

Population mean : 288745.5937364679

Population standard deviation : 291620.51628526766

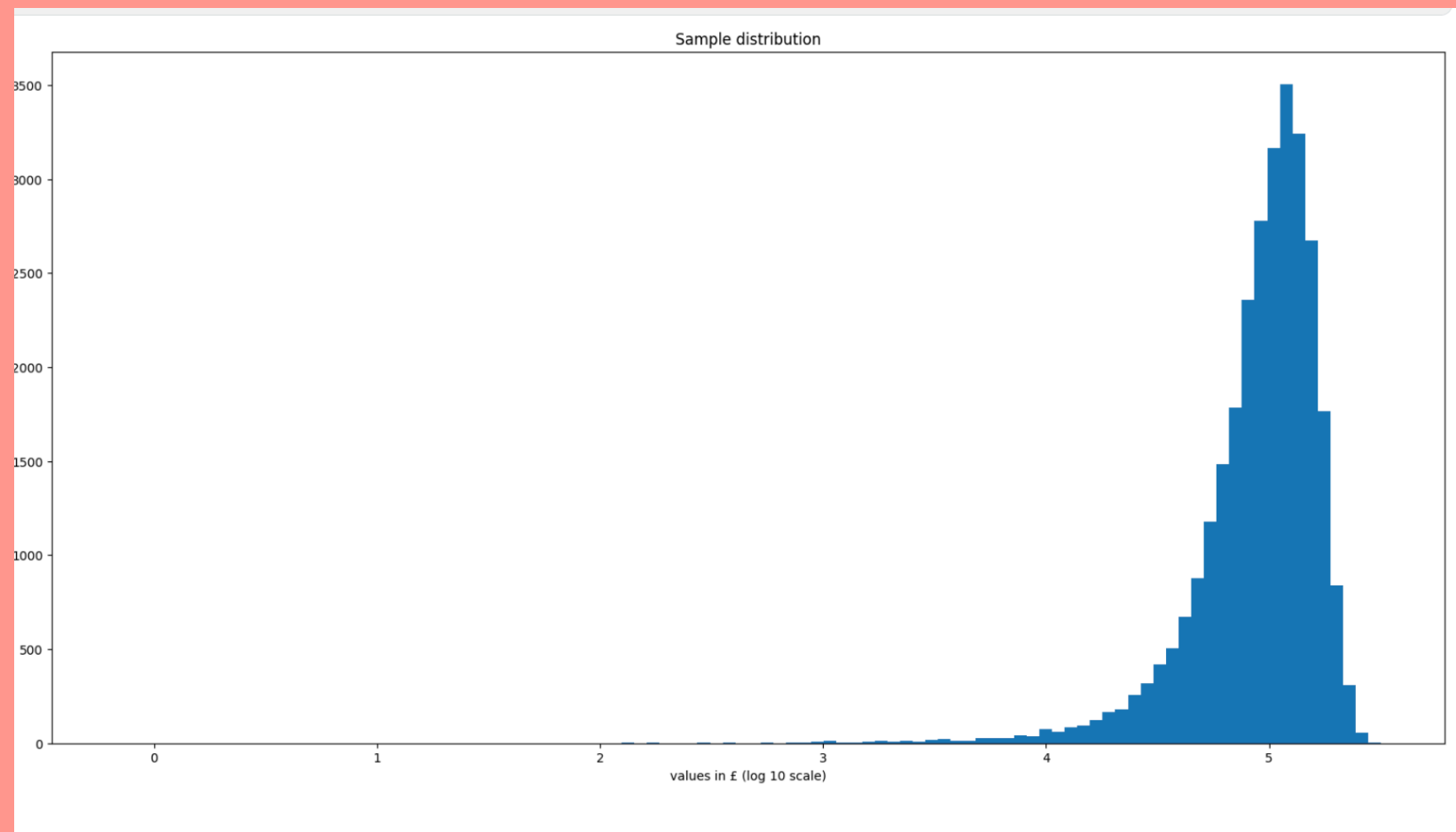
Sample mean : 286791.8382333333

Sample standard deviation : 253636.6435013681



Sample size : 30000 - not a good sample

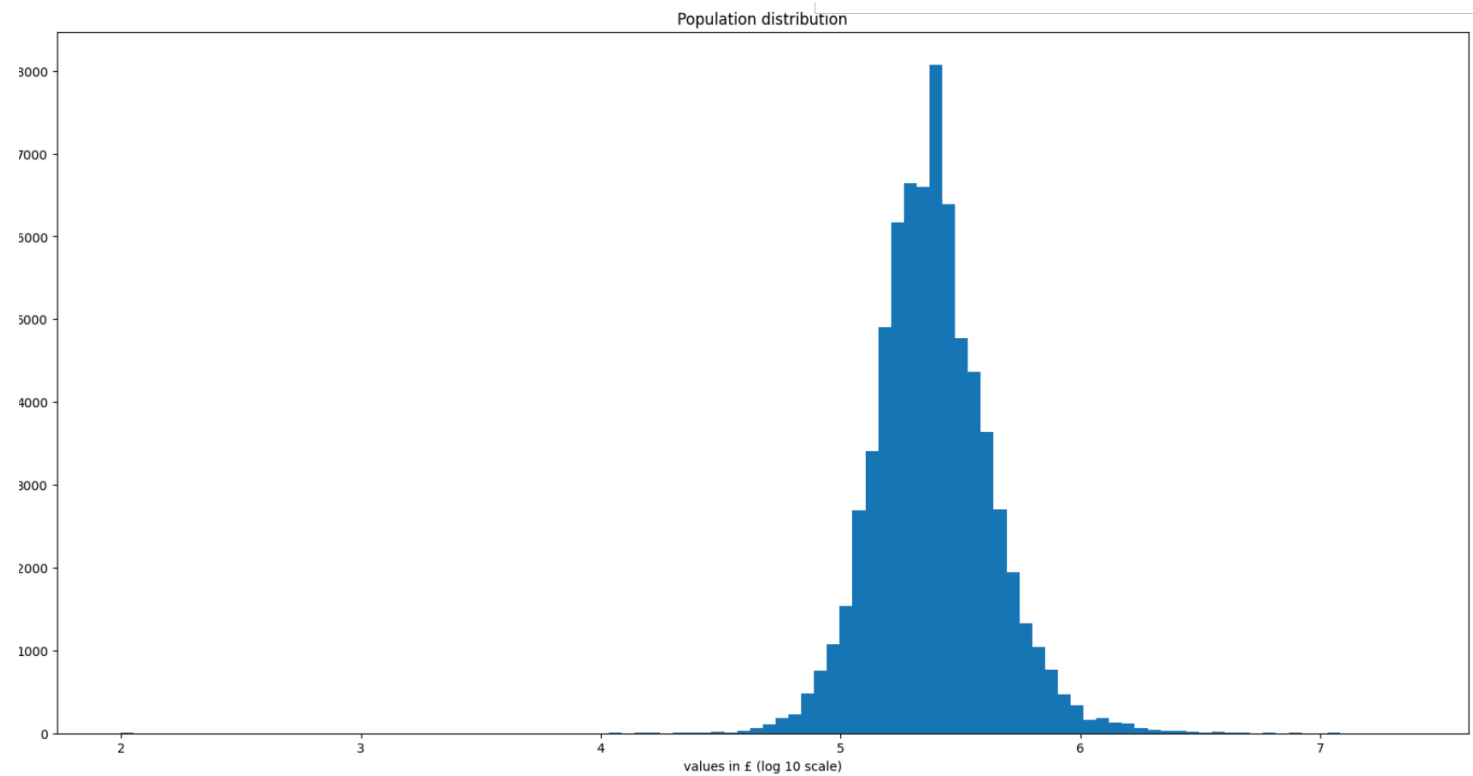
Population mean : 288745.5937364679
Population standard deviation : 291620.51628526766
Sample mean : 100587.60050224914
Sample standard deviation : 50321.46735044537



Sample size effect

Sample size	Effect size	Proportion of the sample against the null
10	0.007	0.01%
100	0.019	0.1%
1000	0.012	1%
10000	-0.010	14%
30000	0.003	42%
30000 (Bad sampling)	764	42%

Comparing the population and the sampling, when its size is 30,000

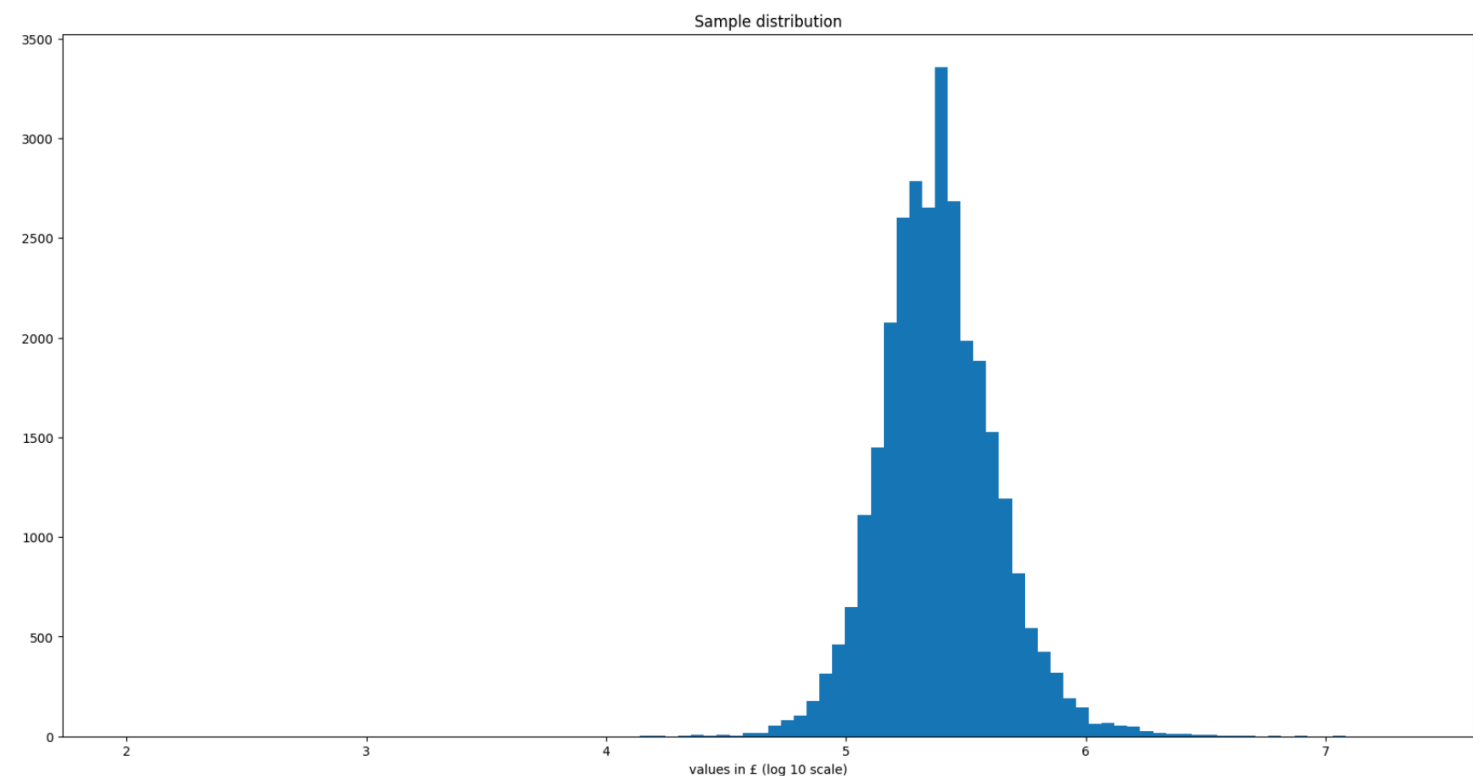


Population mean :

288745.5937364679

Population standard deviation :

291620.51628526766



Sample mean :

286791.8382333333

Sample standard deviation :

253636.6435013681